Causal Inference and ML: From Graphs to Effects

DAGs, Do-Operator, Identification, and Discovery

Kristian Boroz

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Why Causality Matters

Correlation is Not Causation

- Ice cream sales correlate with drowning deaths
- Both are driven by a **confounder**: hot weather
- Acting on correlation alone leads to **wrong decisions**

Why We Need Causal Thinking

- **Prediction:** What will happen?
- **Intervention:** What happens if we *act*?
- **Counterfactual:** What would have happened?
- Causal reasoning answers **interventional** and **counterfactual** questions that pure statistics cannot

Pearl's Ladder of Causation

1. **Association** $P(Y|X)$: Seeing/observing
2. **Intervention** $P(Y|do(X))$: Doing/intervening
3. **Counterfactual** $P(Y_x|X', Y')$: Imagining

The Ladder of Causation in Detail

Level	Activity	Question	Example
1. Association	Observing	What is? If I see X, what is Y?	If someone buys ice cream, will they drown?
2. Intervention	Doing	What if I do X?	What if I force everyone to eat ice cream?
3. Counterfactual	Imagining	What if I had done X instead?	Would this patient have survived if treated differently?

Key Insight

Standard machine learning operates at **Level 1** only. Causal inference is needed to answer **Level 2 and Level 3** questions.

Potential Outcomes Framework

Rubin's Potential Outcomes

For each unit i, define:

- $Y_i(1)$: outcome if unit i receives treatment
- $Y_i(0)$: outcome if unit i receives control
- **Individual Treatment Effect:** $\tau_i = Y_i(1) - Y_i(0)$

The Fundamental Problem:

- We can only ever observe **one** potential outcome per unit
- $Y_i^{obs} = W_i \cdot Y_i(1) + (1 - W_i) \cdot Y_i(0)$
- The other outcome is the **counterfactual** — forever unobserved
- Individual effects τ_i are **never directly identified**

What We Can Identify:

- **ATE:** $\tau = \mathbb{E}[Y_i(1) - Y_i(0)]$
- **CATE:** $\tau(x) = \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i = x]$
- These require **identification assumptions**

Core Identification Assumptions

Assumption 1: Unconfoundedness (Ignorability)

$$\{Y_i(0), Y_i(1)\} \perp\!\!\!\perp W_i \mid X_i$$

Given observed covariates X_i, treatment assignment is **as good as random**.

Assumption 2: Overlap (Positivity)

$$0 < e(x) = P(W_i = 1 \mid X_i = x) < 1 \quad \forall x$$

Every unit has a **non-zero probability** of receiving either treatment or control.

Assumption 3: SUTVA (Stable Unit Treatment Value Assumption)

- **No interference:** Treatment of unit i does not affect outcomes of unit j
- **No hidden versions:** There is only one version of each treatment level

Quiz 1: Identification Assumptions

Which assumption is violated if patients in a clinical trial discuss their treatments with each other and change their behavior accordingly?

1. Unconfoundedness
2. Overlap
3. SUTVA — specifically the no-interference component
4. The exclusion restriction

Quiz 1 Solution

1. Incorrect: Unconfoundedness concerns whether treatment assignment is related to potential outcomes given covariates
2. Incorrect: Overlap concerns whether all units have a positive probability of receiving each treatment
3. **Correct: When patients influence each other's outcomes through discussing treatments, the no-interference component of SUTVA is violated — one unit's treatment affects another's outcome**
4. Incorrect: The exclusion restriction is an instrumental variable concept

Introduction to DAGs

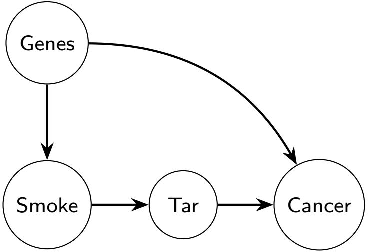
What is a DAG?

- **D**irected **A**cyclic **G**raph
- **Nodes:** Random variables
- **Directed edges ($\rightarrow$):** Direct causal relationships
- **Acyclic:** No variable can cause itself (no feedback loops)
- Encodes **qualitative causal assumptions** about a system

Why Use DAGs?

- Make assumptions **explicit and transparent**
- Determine what to **control for** (and what not to)

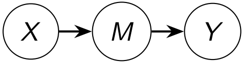
Example DAG: Smoking and Cancer



- Smoking $\rightarrow$ Tar $\rightarrow$ Cancer: **mediation**
- Genes $\rightarrow$ Smoking and Genes $\rightarrow$ Cancer: **confounding**

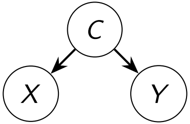
Three Fundamental DAG Structures

1. Chain (Mediation)



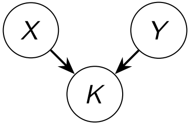
- X and Y are **dependent**
- Controlling for M **blocks** the path
- $X \perp\!\!\!\perp Y \mid M$
- **Don't** control for mediators if you want total effect

2. Fork (Confounding)



- X and Y are **dependent** via C
- Controlling for C **blocks** the path
- $X \perp\!\!\!\perp Y \mid C$
- **Do** control for confounders

3. Collider



- X and Y are **independent**
- Controlling for K **opens** the path!
- $X \not\perp\!\!\!\perp Y \mid K$
- **Never** control for colliders

The Collider Trap

Controlling for a collider introduces **spurious associations** between its parents. This is one of the most common mistakes in empirical research.

D-Separation

Definition: D-Separation

Two sets of nodes X and Y are **d-separated** by a set Z if every path between X and Y is **blocked** by Z. D-separation implies **conditional independence**: $X \perp\!\!\!\perp Y \mid Z$.

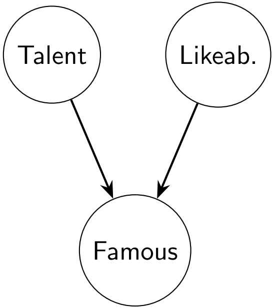
A path is blocked by Z if:

- The path contains a **chain** $A \to B \to C$ or **fork** $A \leftarrow B \to C$ where $B \in Z$ (conditioning blocks the path)
- The path contains a **collider** $A \to B \leftarrow C$ where $B \notin Z$ **and** no descendant of B is in Z (not conditioning keeps it blocked)

Practical Use:

- Use d-separation to determine which variables to **condition on** for identification
- Use it to check whether a proposed adjustment set **blocks all backdoor paths**
- Automated tools (e.g., `dagitty`, `DoWhy`) can compute d-separation programmatically

Collider Bias: The Hollywood Example



- In the general population: Talent and Likability are **independent**
- **Fame** is a **collider** — a common effect of both
- Once we condition on Fame (e.g., study only famous people): a **spurious negative correlation** appears
- *"You don't need talent if you're likeable enough"*

The Collider Rule

Conditioning on a **common effect** of two independent variables makes them appear dependent. Never control for colliders.

Collider Bias: Real Research Examples

Berkson's Paradox

- Studying hospital patients only
- Hospitalization is a **collider**
- Two unrelated conditions appear **negatively correlated** within the hospital sample

Birth Weight Paradox

- Controlling for birth weight **reverses** the apparent effect of smoking on infant mortality
- Birth weight is a **collider** between smoking and other risk factors

Selection Bias in Surveys

- Conditioning on survey response induces **spurious correlations**
- Survey response is a **collider** between variables of interest

Common Pattern Across All Examples:

- A **selection** or **conditioning** step creates the collider
- The resulting bias can **flip signs** or **create associations** from nothing
- Often mistaken for a real causal finding

The Do-Operator

Definition

The **do-operator** $do(X = x)$ represents a **physical intervention** that sets X to value x, regardless of what would naturally determine X.

$$P(Y \mid do(X = x)) \neq P(Y \mid X = x) \quad \text{in general}$$

Observational vs. Interventional:

- $P(Y \mid X = x)$: Distribution of Y among units **observed** to have $X = x$
- $P(Y \mid do(X = x))$: Distribution of Y if we **forcibly set** $X = x$ for everyone
- These differ whenever X has **confounders**

What Do Does Graphically:

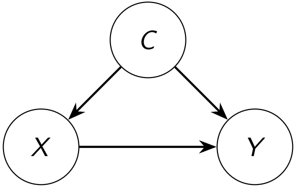
- Removes all **incoming edges** to X
- Creates a **mutilated graph** where X is set externally
- Eliminates confounder influence on X
- Equivalent to a **randomized experiment**

The Do-Operator: Connection to Potential Outcomes

Key Identity:

$$P(Y \mid do(W = w)) = P(Y(w))$$

- The do-operator and the **potential outcomes framework** are two languages for the same concept
- $Y(w)$ is the outcome we *would* observe if treatment were set to w
- Both frameworks aim to answer: *what is the effect of acting on W?*



Observing $X = x$: includes confounder C

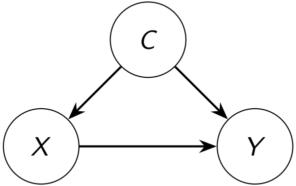
Doing $do(X = x)$: removes arrow $C \to X$

The Backdoor Criterion

Definition

A set Z satisfies the **backdoor criterion** relative to (X, Y) if:

1. Z **blocks all backdoor paths** from X to Y
2. Z contains **no descendants** of X



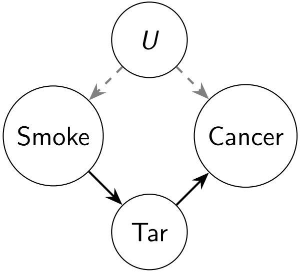
- Backdoor path: $X \leftarrow C \rightarrow Y$
- $Z = \{C\}$ blocks this path and is not a descendant of X
- **Adjustment formula:**

$$P(Y \mid do(X)) = \sum_c P(Y \mid X, C = c)P(C =$$

The Frontdoor Criterion

A set M satisfies the **frontdoor criterion** relative to (X, Y) if:

1. All directed paths from X to Y pass **through** M
2. No unblocked backdoor paths from X to M
3. All backdoor paths from M to Y are **blocked by** X



- U (genetics) is **unobserved** — backdoor criterion fails
- $M = \{\text{Tar}\}$ satisfies the frontdoor criterion
- **Frontdoor adjustment:**

$$P(C \mid do(S)) = \sum_t P(t \mid S) \sum_s P(C \mid s, t) P(s)$$

Quiz 2: DAGs and Identification

In a DAG where U is an unmeasured confounder affecting both X and Y, and M is a measured mediator on the only path from X to Y, which criterion can identify $P(Y \mid do(X))$?

1. The backdoor criterion using $\{U\}$
2. The backdoor criterion using $\{M\}$
3. The frontdoor criterion using $\{M\}$
4. Neither criterion applies — the effect is not identified

Quiz 2 Solution

1. Incorrect: U is **unmeasured** and cannot be conditioned on
2. Incorrect: Conditioning on the mediator M would **block the causal path** we want to measure, and M is a descendant of X — violating the backdoor criterion
3. **Correct: The frontdoor criterion applies. Since all paths go through M, there are no unblocked backdoor paths from X to M, and X blocks all backdoor paths from M to Y**
4. Incorrect: The frontdoor criterion provides identification in this case

The Propensity Score

Definition

The **propensity score** is the probability of receiving treatment given observed covariates:

$$e(x) = P(W_i = 1 \mid X_i = x)$$

Key Theorem (Rosenbaum & Rubin, 1983):

- If unconfoundedness holds given X:

$$\{Y(0), Y(1)\} \perp\!\!\!\perp W \mid X$$

- Then it also holds given $e(X)$:

$$\{Y(0), Y(1)\} \perp\!\!\!\perp W \mid e(X)$$

Why This Matters:

- Reduces a **high-dimensional** adjustment problem to a **one-dimensional** scalar
- Enables matching, stratification, and weighting on $e(x)$ alone
- Estimated in practice via **logistic regression** or ML classifiers
- Always check **overlap** by plotting $\hat{e}(x)$ for treated vs. control

Propensity Score Methods: Matching and Stratification

1. **Matching**
   - For each treated unit, find the **most similar** control unit based on $\hat{e}(x)$
   - Creates a **pseudo-experiment** where groups are comparable
   - **Variants:** 1-to-1, k-nearest neighbors, caliper matching
   - **Limitation:** Discards unmatched units — reduces sample size

2. **Stratification**
   - Divide units into **bins** based on $\hat{e}(x)$ quintiles
   - Estimate treatment effects **within each bin**
   - Average across bins to obtain the overall ATE
   - **Advantage:** Uses all observations, no units discarded

Propensity Score Methods: Weighting

Inverse Propensity Weighting (IPW)

$$\hat{\tau}_{IPW} = \frac{1}{n} \sum_i \left(\frac{W_i Y_i}{\hat{e}(X_i)} - \frac{(1 - W_i) Y_i}{1 - \hat{e}(X_i)} \right)$$

Intuition:

- Weight each unit by the **inverse probability** of receiving its actual treatment
- Creates a **pseudo-population** where treatment is independent of covariates
- Upweights **underrepresented** units in each treatment arm

Known Issue:

- **Extreme weights** when $\hat{e}(x) \approx 0$ or $\hat{e}(x) \approx 1$
- Can cause high variance in estimates
- **Fix:** Trim or clip extreme weights

Doubly Robust Estimation (AIPW)

Augmented IPW Estimator

$$\hat{\tau}_{AIPW} = \frac{1}{n}\sum_i \left[\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i) + \frac{W_i(Y_i - \hat{\mu}_1(X_i))}{\hat{e}(X_i)} - \frac{(1-W_i)(Y_i - \hat{\mu}_0(X_i))}{1-\hat{e}(X_i)}\right]$$

Key Properties:

- Combines **outcome regression** $\hat{\mu}_w(x)$ with **IPW**
- **Doubly robust:** Consistent if *either* the outcome model or the propensity model is correctly specified
- More **efficient** than IPW alone

Intuition:

- Outcome model provides a **baseline prediction**
- IPW term **corrects residual bias** if the outcome model is wrong
- If both models are correct: achieves **semiparametric efficiency bound**

Quiz 3: Propensity Scores

What is the consequence of having propensity scores very close to 0 or 1 for some units?

1. It reduces the variance of IPW estimates
2. It indicates poor overlap, leading to extreme IPW weights and unreliable causal estimates
3. It triggers the backdoor criterion automatically
4. It makes AIPW estimates inconsistent

Quiz 3 Solution

1. Incorrect: Extreme propensity scores cause **high** variance, not low
2. **Correct: Propensity scores near 0 or 1 indicate poor overlap. Units with extreme scores receive extreme IPW weights $\frac{1}{\hat{e}(x)}$ or $\frac{1}{1-\hat{e}(x)}$, making estimates highly sensitive to those observations**
3. Incorrect: The backdoor criterion is a graphical concept unrelated to propensity score values
4. Incorrect: AIPW is more robust but extreme overlap violations affect all estimators

Causal Discovery: Learning the Graph from Data

What is Causal Discovery?

- So far we **assumed** the DAG is known
- Causal discovery: **learn** the causal structure from data
- Combines **statistical testing** with **causal assumptions**
- Output: A DAG or equivalence class of DAGs (CPDAG)

Key Challenge:

- Many DAGs imply the **same conditional independences**
- These form a **Markov Equivalence Class (MEC)**

Main Approaches:

1. **Constraint-Based:**
   - Test conditional independences
   - Use d-separation to infer edges
   - Examples: PC algorithm, FCI
2. **Score-Based:**
   - Score each DAG (e.g., BIC)
   - Search over DAG space
   - Examples: GES algorithm
3. **Functional Causal Models:**
   - Assume structural equation form
   - Use noise asymmetry to orient edges
   - Examples: LiNGAM, ANM

The PC Algorithm

PC Algorithm Steps

1. **Start** with a complete undirected graph over all variables
2. **Remove edges** between variable pairs that are conditionally independent given some subset of other variables
3. **Orient v-structures (colliders):** If $X - Z - Y$, X and Y are not adjacent, and $X \not\perp\!\!\!\perp Y \mid Z$, orient as $X \to Z \leftarrow Y$
4. **Propagate orientations** using Meek rules to orient remaining undirected edges without creating new v-structures or directed cycles

Strengths:

- Theoretically sound under faithfulness assumption
- Scales reasonably to moderate dimensions
- Outputs a **CPDAG** representing the

Limitations:

- Sensitive to errors in conditional independence tests
- Cannot distinguish all edges within an MEC
- Requires **faithfulness assumption**

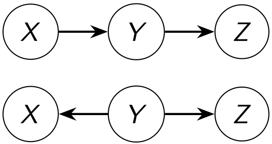
Markov Equivalence Classes

Markov Equivalence

Two DAGs are **Markov equivalent** if and only if they share the same:

- **Skeleton** (same edges, ignoring direction)
- **V-structures** (same colliders $X \to Z \leftarrow Y$)

Example: Equivalent DAGs



Both imply the **same** conditional independences

What Breaks Equivalence?

- **Interventional data:** Intervening on X distinguishes $X \to Y$ from $X \leftarrow Y$
- **Non-Gaussian noise:** LiNGAM uses distributional asymmetry to orient edges
- **Temporal ordering:** If X precedes Y in time, direction is resolved
- **Domain knowledge:** Expert constraints can fix edge orientations

Functional Causal Models: LiNGAM = Linear Non-Gaussian Acyclic Model

Assume data is generated by:

$$X_j = \sum_{k \in \mathrm{pa}(j)} b_{jk} X_k + \varepsilon_j, \qquad \varepsilon_j \sim \text{non-Gaussian, mutually independent}$$

Key Insight:

- Under **Gaussian** noise: $X \to Y$ and $X \leftarrow Y$ are observationally equivalent
- Under **non-Gaussian** noise: the causal direction is **fully identifiable** from observational data
- Uses **Independent Component Analysis (ICA)** to recover the full DAG

Practical Implications:

- **Full identifiability:** No equivalence class — a unique DAG is recovered
- **Limitation:** Requires genuinely non-Gaussian errors
- **Extensions:** Post-nonlinear models, Additive Noise Models (ANM) for nonlinear settings
- Useful when **randomization is impossible** but distributional

Quiz 4: Causal Discovery

A researcher runs the PC algorithm and finds that the edge between A and B cannot be oriented. What does this most likely mean?

1. There is definitely no causal relationship between A and B
2. Both $A \rightarrow B$ and $A \leftarrow B$ are consistent with the observed conditional independences — they belong to the same Markov Equivalence Class
3. The algorithm has failed and should be re-run with different parameters
4. A and B must be conditionally independent given all other variables

Quiz 4 Solution

1. Incorrect: An undirected edge means a relationship **exists** but its direction cannot be determined from observational data alone
2. **Correct: An unoriented edge in the CPDAG output means both directions are Markov equivalent. Observational data alone cannot distinguish between them — interventional data or additional assumptions are needed**
3. Incorrect: This is **expected and correct** output — the PC algorithm correctly reports the MEC when edge directions are not identified
4. Incorrect: If A and B were conditionally independent, the edge between them would have been **removed entirely** in Step 2 of the algorithm

A Complete Causal Workflow - Step-by-Step Causal Analysis

1. **Define the causal question:** What is the treatment W? What is the outcome Y?
2. **Draw the DAG:** Encode domain knowledge about all relevant variable relationships
3. **Check identification:** Does the backdoor or frontdoor criterion hold?
4. **Select adjustment set:** Use the minimal set satisfying the relevant criterion
5. **Estimate the effect:** Apply regression, IPW, AIPW, or matching estimators
6. **Validate assumptions:** Overlap plots, placebo tests, sensitivity analysis
7. **Sensitivity analysis:** How robust are conclusions to unmeasured confounders?

Common Mistakes

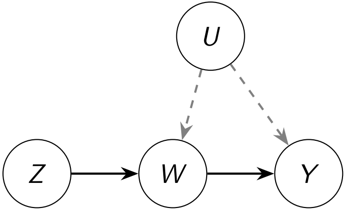
- **Controlling for mediators:** Blocks the causal pathway you want to measure (post-treatment bias)
- **Controlling for colliders:** Opens spurious non-causal associations
- **Omitting confounders:** Leaves backdoor paths open, biasing estimates

Instrumental Variables (IV)

What Makes a Valid Instrument?

A variable Z is a valid **instrument** for the effect of W on Y if:

1. **Relevance:** Z is correlated with treatment W: $\text{Cov}(Z, W) \neq 0$
2. **Exclusion:** Z affects Y *only through* W — no direct path $Z \to Y$
3. **Exogeneity:** Z is independent of all unmeasured confounders of W and Y



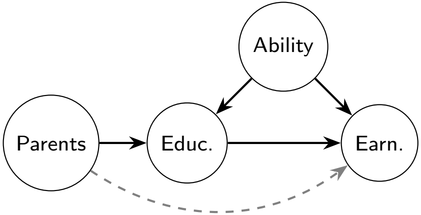
Two-Stage Least Squares (2SLS):

- **Stage 1:** Regress W on Z, obtain $\hat{W}$
- **Stage 2:** Regress Y on $\hat{W}$:

$$\hat{\tau}_{IV} = \frac{\text{Cov}(Y, Z)}{\text{Cov}(W, Z)}$$

Worked Example: Education and Earnings

Causal DAG:



The Problem:

- Backdoor paths via **Ability** and **Parents**
- Ability is typically **unobserved** — backdoor adjustment infeasible

IV Solution: Distance to College

- **Relevant:** Predicts years of education
- **Excludable:** No direct effect on earnings
- **Exogenous:** Unrelated to ability or family background
- **Result:** Identifies causal effect of education on earnings despite unobserved confounding

Quiz 5: Complete Causal Workflow

A researcher wants to estimate the effect of a job training program (W) on employment (Y). They observe that participants self-select into the program. Which of the following represents the correct first step?

1. Immediately run an OLS regression of Y on W
2. Draw a DAG encoding all known causes of both program participation and employment, then check identification
3. Apply IPW weighting using estimated propensity scores without further checks
4. Use LiNGAM to discover the causal graph automatically

Quiz 5 Solution

1. Incorrect: Self-selection means OLS will be confounded — participants may differ systematically from non-participants in ways that also affect employment
2. **Correct: Drawing a DAG first makes all causal assumptions explicit, identifies confounders, detects potential colliders, and determines whether identification is even possible before choosing an estimator**
3. Incorrect: IPW requires first verifying that unconfoundedness holds given observed covariates — this requires domain knowledge encoded in a DAG
4. Incorrect: Automated discovery is a supplement to, not a replacement for, domain knowledge — and may not orient all edges

From Causal Graphs to CATE Estimation

DAGs Tell Us What to Adjust For:

- Backdoor criterion identifies the correct adjustment set X
- Backdoor adjustment gives us:

$$\tau = \mathbb{E}\big[\mathbb{E}[Y|W{=}1, X] - \mathbb{E}[Y|W{=}0, X]\big]$$

- This is exactly what **S-, T-, X-, and R-learners** estimate
- DAGs justify **which** covariates to include — and which to exclude

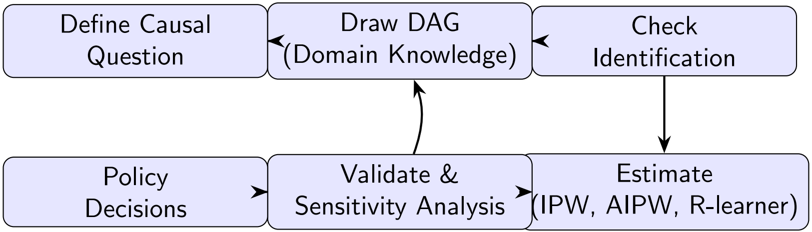
From DAG Assumptions to R-Loss:

- Robinson's decomposition:

$$Y - m(X) = (W - e(X))\,\tau(X) + \varepsilon$$

- $m(X)$ and $e(X)$ are the **nuisance functions** implied by the DAG's adjustment set
- **Unconfoundedness** (from DAG) justifies using observational data
- **Overlap** ensures $e(x) \in (0, 1)$ is well-defined

Summary: The Causal Inference Pipeline



- The pipeline is **iterative**: validation may require revisiting the DAG
- **Domain knowledge** is indispensable — data alone cannot establish causality
- **Modern ML** provides powerful estimators once identification is established

Key Takeaways: Causal Inference

- **Pearl's Ladder** distinguishes association, intervention, and counterfactual reasoning — standard ML only operates at the first level
- **DAGs** make causal assumptions explicit and enable graphical identification via backdoor and frontdoor criteria
- **D-separation** provides a systematic way to determine conditional independence and valid adjustment sets
- **Collider bias** is a critical and often overlooked pitfall — never condition on common effects of treatment and outcome
- **Propensity scores** reduce confounding adjustment to a one-dimensional problem but require overlap
- **Causal discovery** algorithms (PC, LiNGAM) can learn graph structure from data under assumptions, but domain knowledge remains essential
- **Identification assumptions** (unconfoundedness, overlap, SUTVA) are the bridge between observational data and causal claims

Summary Table: Causal Inference Methods

Method	Key Idea	Strengths	Limitations
Backdoor Adj.	Control for observed confounders	Simple, interpretable	Fails with unmeasured confounders
Frontdoor Adj.	Adjust via measured mediator	Works with hidden confounders	Requires full mediation
IPW	Reweight by propensity score	Targets ATE directly	High variance with extreme weights
AIPW	IPW + outcome regression	Doubly robust, efficient	Requires both nuisance models
IV / 2SLS	Exploit exogenous variation	Handles unmeasured confounders	Requires valid instrument
PC Algorithm	CI testing on observational data	No parametric assumptions	Returns MEC, not unique DAG
LiNGAM	Non-Gaussian noise model	Full DAG identifiability	Requires non-Gaussian errors